Advanced Mathematical Techniques Worksheet 3
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The purpose of this worksheet is to give you some more practice with Dirichlet series and infinite products, then working with the various representations of the Gamma function. Please show all of your work without a calculator, but also please check your results using Mathematica or another computer algebra system.

1. The cosine function has the infinite product representation $\cos x = \prod_{k=0}^{\infty} \left[1 - \frac{4x^2}{(2k+1)^2 \pi^2} \right]$.

   (a) Explain why this product converges for all x except those for which it is zero.

   (b) Take the logarithm of this product expansion and expand each of the logarithms to get a Maclaurin series for $\ln \cos x$. Express the coefficients of this expansion in terms of a Dirichlet series.

   (c) Differentiate your result from part (b) to determine a Maclaurin series for another well-known trigonometric function in terms of Dirichlet series.

2. This problem concerns the Maclaurin series for the logarithm of the Gamma function,

$$\ln \Gamma(1+z) = -\gamma z + \sum_{k=2}^{\infty} \frac{(-1)^k \zeta(k)}{k} z^k; \ |z| < 1.$$

   (a) Substitute $z = -1/2$ into this equation to determine an interesting relationship between π and the values of the zeta function. Does your resulting series converge absolutely or conditionally?

   (b) Substitute $z = 1$ into this equation to determine an interesting relationship between γ and the values of the zeta function. Does your resulting series converge absolutely or conditionally?

3. Use the recurrence relation for the Gamma function to determine the value of $\Gamma(13/3)$ in terms of $\Gamma(1/3)$.

4. Use the integral representation of the Gamma function to determine the value of the integral $\int_0^{\infty} x^2 \sqrt{x}\, e^{-2x}\, dx$. You may include various rational number powers of 2 and/or π in your result.

5. Use the integral representation of the Gamma function to determine the value of the integral $\int_0^\infty x\, e^{-2x^3}\, dx$. You may include a value of the Gamma function evaluated between 0 and $1/2$ in your answer, along with various rational number powers of 2 and/or π.

6. This problem concerns products of the form $\prod_{n=1}^{\infty}\left(1 \pm \frac{x}{n}\right)$.

   (a) Write out the first five terms in the product $P_N = \prod_{n=1}^{N}\left(1+\frac{1}{n}\right)$ as improper fractions in order to find a pattern to determine a closed-form expression for P_N. What happens as *N* tends to infinity?

   (b) Write out the first five terms in the product $Q_N = \prod_{n=2}^{N}\left(1-\frac{1}{n}\right)$ as fractions in order to find a pattern to determine a closed-form expression for Q_N. What happens as *N* tends to infinity?

   (c) Explain why the product given at the beginning of this problem should be understood as *divergent* for all values of *x*.

   (d) Explain why the product $A = \prod_{n=1}^{\infty}\left[\left(1+\frac{x}{n}\right)e^{-x/n}\right]$ *is* convergent for all *x* except negative integers even though the original product in this problem diverges.

   (e) Explain why the product $B = \prod_{n=1}^{\infty}\left[\left(1+\frac{x}{n}\right)\ln\left(\frac{e}{1+x/n}\right)\right]$ *is* convergent for all positive numbers *x* even though the original product in this problem diverges. Would this product converge if the *e* was replaced with a 2? Explain why or why not.

   (f) Explain why part (d) may be a better choice to provide an expansion that converges than (e). It might be good to think about the differences between where these products are defined.